ConvPEFT: Adapting Transformer PEFT to CNN-based Audio Classification

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Models and Codes are publicly available:
Our Team's github repogitory

Introduction

- Audio classification is crucial for environmental sound monitoring, security systems, and smart devices
- Deep learning models (CNNs) achieve high accuracy but require extensive fine-tuning for new domains
- Parameter-Efficient Fine-Tuning (PEFT) methods reduce computational costs while maintaining performance
- Challenge: PEFT methods designed for Transformers need adaptation for CNN architectures

PEFT Methods

LoRA (Low-Rank Adaptation)

- Originally for Transformer attention layers
- Injects trainable low-rank matrices (A, B)

$$W_0 \in \mathbb{R}^{d\times k},\ A \in \mathbb{R}^{r\times k},\ B \in \mathbb{R}^{d\times r} \ll min(d,k)$$

Adaptive Adapter

- Sequential bottleneck layers
- Down-projection → Non-linearity → Up-projection

$$W_{down} \in \mathbb{R}^{d\times r},\ W_{up} \in \mathbb{R}^{r\times d},\ f = \mathrm{ReLU}$$

Research Objective

Our Contributions:

- Systematic comparison of LoRA and Adapter in CNN-based audio classification
- Novel adaptation strategy for applying PEFT to convolutional layers (4D → 2D)
- Analysis of structural impact on performance

Experiment setup

Dataset: UrbanSound8K

- 8,732 audio clips across 10 urban sound classes(≤4 seconds each)
- 10-fold split strategy prevents data leakage by ensuring slices from the same original recording stay within a single fold
- Purpose: Sound alert system for hearing-impaired users in on-device environments
- Preprocessing: Audio → MFCC spectrograms (40×174×1) for CNN input

Model Architecture

- Base CNN: 3-convolutional-layer architecture for audio classification
- Input: MFCC spectrograms (40×174×1)
- All base convolutional layers (Conv1-3) are frozen; PEFT modules are added to Conv2-3
  - **LoRA**: Parallel low-rank adaptation (A: 3×3, B: 1×1)
  - **Adapter**: Sequential bottleneck with ReLU non-linearity

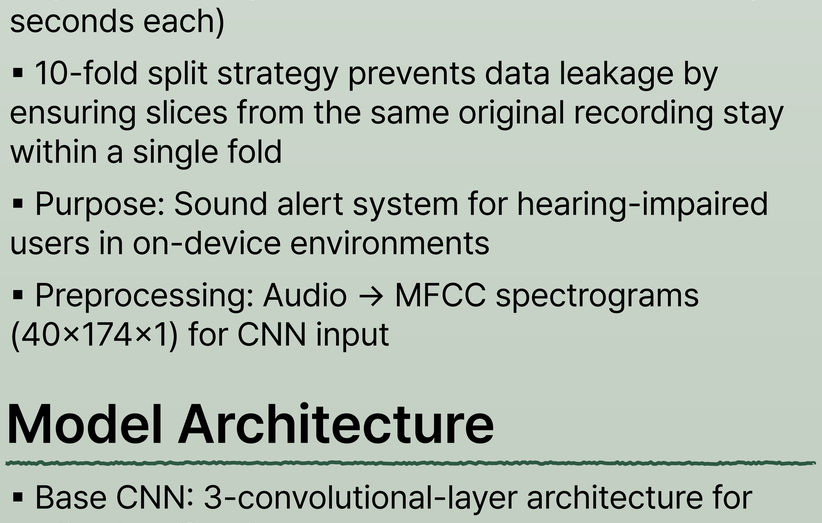

Figure I. CNN Architecture with PEFT Module Integration (Full Fine-Tuning vs Conv-LoRA vs Conv-Adapter)

Method: Conv-PEFT

Conv-LoRA

- Adapts LoRA's low-rank decomposition to Conv2D layers
- Matrix A: 3×3 conv preserves spatial information
- Matrix B: 1×1 pointwise conv for channel projection
- Parallel path: ΔW(x) = Conv_B(Conv_A(x))

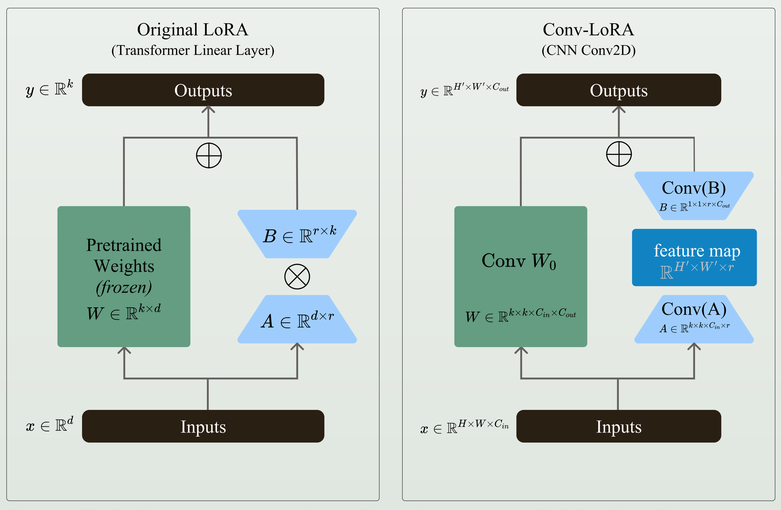

Figure II. LoRA Adaptation from Transformers to CNNs. Our Conv-LoRA preserves spatial information in matrix A while using pointwise convolution in B for efficient channel mixing.

Conv-Adapter

- Adapts sequential bottleneck structure to Conv2D layers
- Down-project → ReLU → Up-project with residual connection
- Serial insertion after main convolution layer
- Non-linearity enables richer feature transformation

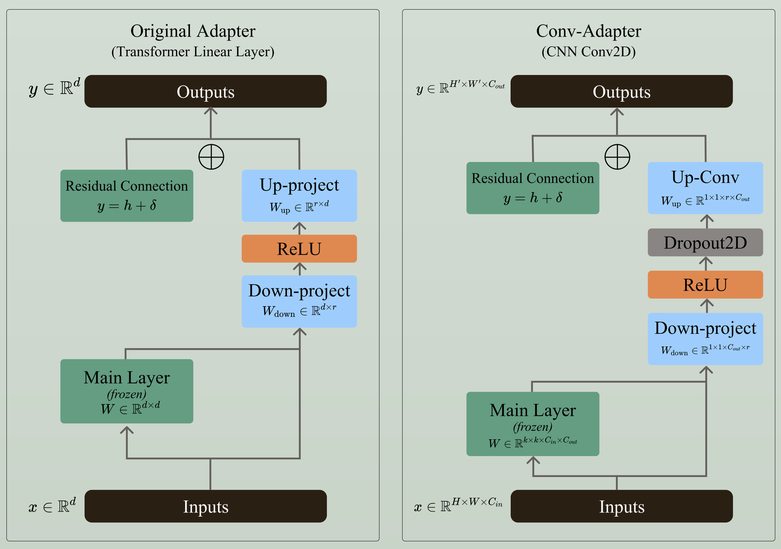

Figure III. Adapter Adaptation from Transformers to CNNs. Our Conv-Adapter uses sequential bottleneck structure with down-projection, ReLU, and up-projection.

Ablation Study

Conv-LoRA

Hyperparameter	Value	Trainable	Accuracy	Loss
Learning Rate				
	1×10^{-3}	26,250	60.10%	1.2853
	$\mathbf{1 \times 10^{-4}}$	26,250	**60.69%**	**1.2007**
	1×10^{-5}	26,250	54.96%	1.2586
Dropout (p)				
	0.4	26,250	**60.69%**	**1.2007**
	0.3	26,250	59.98%	1.1937
	0.2	26,250	58.66%	1.2097
Rank and Alpha				
	$\mathbf{r = 8, \alpha = 16}$	26,250	**60.69%**	**1.2007**
	$r = 4, \alpha = 8$	22,026	58.78%	1.2073
	$r = 16, \alpha = 32$	34,698	60.57%	1.2101

Conv-Adapter

Hyperparameter	Value	Trainable	Accuracy	Loss
Learning Rate				
	5×10^{-5}	38,570	54.84%	1.2357
	1×10^{-4}	38,570	58.54%	1.1979
	$\mathbf{2 \times 10^{-4}}$	38,570	**60.93%**	**1.1694**
Bottleneck Ratio				
	1	59,146	61.77%	1.2254
	2	38,570	**60.93%**	**1.1694**
	4	28,282	59.02%	1.1879
	8	23,138	55.79%	1.2564
Dropout Rate				
	0.1	38,570	**60.93%**	**1.1953**
	0.3 (default)	38,570	60.93%	1.1694
Training Epochs				
	30	38,570	60.93%	1.1694
	50	38,570	**61.17%**	**1.1610**

Result

Conv-LoRA

- Best config: lr=1×10⁻⁴, weight_decay=1×10⁻⁴, rank=8, α=16, dropout=0.4
- Achieves 60.93% accuracy with 26,250 trainable params
- Stable convergence with reduced overfitting vs Full FT
- 76% parameter reduction while improving accuracy

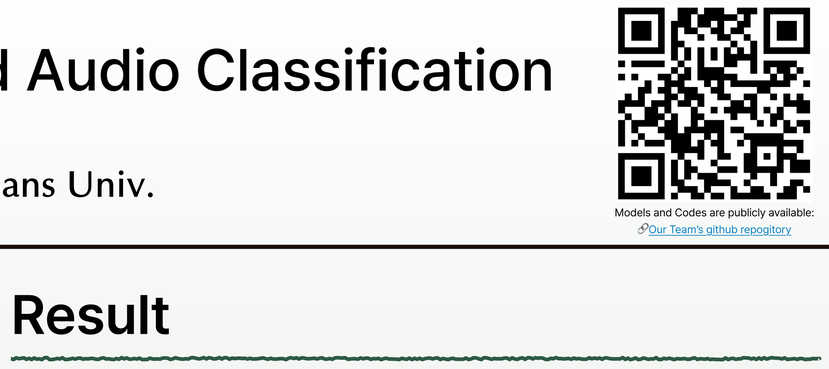

Figure IV. Conv-LoRA Training Curves: Loss and Validation Accuracy across Epochs

Conv-Adapter

- Best config: lr=2×10⁻⁴, ratio=2, dropout=0.1
- Achieves 60.93% accuracy with 38,570 trainable params
- Smooth training curves indicate good generalization
- Non-linear bottleneck provides effective adaptation

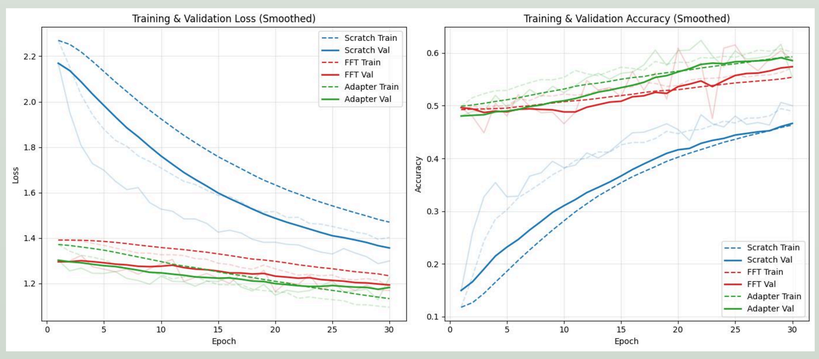

Figure V. Conv-Adapter Training Curves: Loss and Validation Accuracy across Epochs

Comparison

Method	Params	Accuracy	Loss
Scratch	110,474 (100%)	52.45%	1.287
Full FT	110,474 (100%)	59.38%	1.216
LoRA	26,250 (23.8%)	60.69%	1.201
Adapter	38,570 (34.9%)	**62.01%**	1.227

Analysis & Conclusion

✓ Adapter achieves the best accuracy (62.01%), and LoRA provides the best parameter efficiency (23.8%)

✓ Both PEFT methods outperform Full Fine-Tuning despite using far fewer parameters

✓ PEFT enables lightweight, high-performance on-device audio model adaptation

Future Works

- Model Scaling: Evaluate conv-PEFT on larger model architectures
- Dataset Expansion: Test our method on broader audio datasets beyond UrbanSound8k
- Hybrid&Deployment: Explore LoRA-Adapter hybrids and assess on-device feasibility

References

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